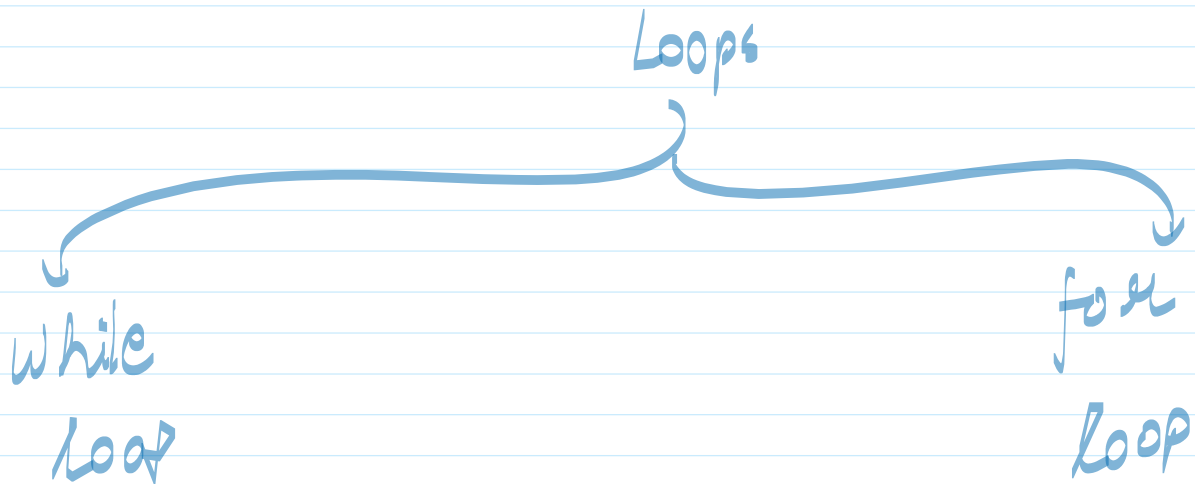


Loops:-

- Wake up
- Brush
- Eat Breakfast

Repeats Everyday

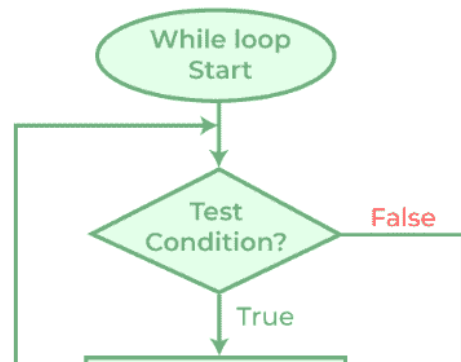
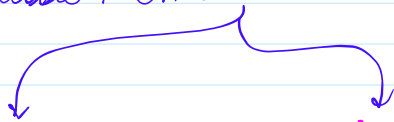
loop

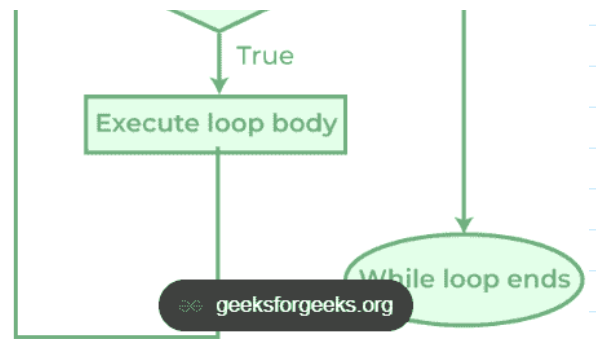
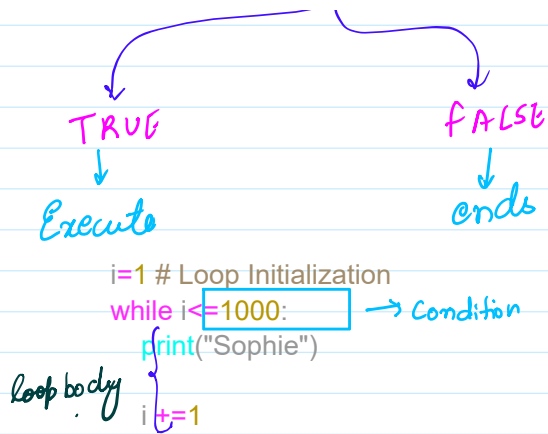


While loop:-

1) Loop start (Initialization)

2) Condition checks





$i = 7 \neq 1000$

$2 \leq 1000 \rightarrow \text{True}$

$1001 \leq 1000 \rightarrow \text{false}$

1. Sophie

2. Sophie

3. Sophie

Iteration:-

➤ One cycle of success loop is called iteration.

While loop with Else:-

```

condition=False
while condition==True:
    print("Sophie")
else:
    print("Azaan")
  
```

Range function :-

range(start : stop: step)

range(0, 5): 0 1 2 3 4

range(1, 11): 1 2 3 4 5 6 7 8 9 10

Observation: Last no. does not count

For loop :-

```
for i in range(1, 1001):  
    print("Sophie")
```

i=1 # Loop Initialization

while i <= 1000:

→ print("Sophie")

i += 1

automatic

```
for i in range(1, 11, 2):  
    print("Sophie")
```

○

✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓
1 2 3 4 5 6 7 8 9 10
i ↪ ↪ ↪ ↪ ↪ ↪ ↪ ↪ ↪

Sum of N natural number :

Ex- $1 + 2 + 3 + 4 + 5 + \dots + N = \frac{N(N+1)}{2}$

$N = 5$

Sum = 0  Initially.

while